



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

## Faculty of Natural and Agricultural Sciences

Fakulteit Natuur- en Landbouwetenskappe  
Lefapha la Disaense tša Tlhago le Temo

# Study Guide

## Department of Physics

### Faculty of Natural and Agricultural Sciences

### General Physics for Engineers

**FSK 116**

**2021 Version 2.0**

# Table of Contents

|          |                                                      |                              |
|----------|------------------------------------------------------|------------------------------|
| <b>1</b> | <b>Welcome .....</b>                                 | <b>1</b>                     |
| <b>2</b> | <b>Administrative information.....</b>               | <b>2</b>                     |
| 2.1      | Contact details of the team .....                    | 2                            |
| 2.2      | Prescribed Book .....                                | 3                            |
| 2.3      | Mode of Presentation and Online sessions times ..... | 3                            |
| 2.4      | ClickUP and Cengage's WebAssign .....                | 4                            |
| 2.5      | Important dates .....                                | 5                            |
| 2.6      | Lecture Timetable .....                              | 5                            |
| 2.7      | Code of conduct .....                                | 6                            |
| 2.7.1    | Communication via email .....                        | 6                            |
| 2.7.2    | Plagiarism .....                                     | 6                            |
| 2.7.3    | Compliments and complaints .....                     | 6                            |
| 2.8      | Main Outcomes.....                                   | 7                            |
| 2.9      | Learning assumed to be in place .....                | 7                            |
| 2.10     | Summary of topics and chapters to be covered .....   | 7                            |
| <b>3</b> | <b>Assessment .....</b>                              | <b>8</b>                     |
| 3.1      | Assessment Policy .....                              | 8                            |
| 3.1.1    | General .....                                        | 8                            |
| 3.1.2    | Assessment Approach.....                             | 9                            |
| 3.1.3    | Assessment Opportunities.....                        | 9                            |
| 3.1.4    | Absence from a scheduled Assessment .....            | 9                            |
| 3.1.5    | Examinations .....                                   | 10                           |
| 3.1.6    | Cheat Sheets* .....                                  | 11                           |
| 3.1.7    | Rules for Cheat Sheets .....                         | 11                           |
| 3.1.8    | Pass Mark .....                                      | 11                           |
| 3.1.9    | Grading and administrative errors .....              | 11                           |
| 3.1.10   | Barcode Stickers.....                                | Error! Bookmark not defined. |
| 3.1.11   | Practical component of the course .....              | 12                           |
| 3.1.12   | Homework and Tutorial Sessions.....                  | 12                           |
| 3.2      | Calculation of the Final Mark (Pass Mark) .....      | 12                           |
| <b>4</b> | <b>Support services .....</b>                        | <b>13</b>                    |
| 4.1      | Safety in the evening and emergencies .....          | 13                           |
| 4.2      | E-learning support.....                              | 13                           |
| 4.3      | Other support services: .....                        | 13                           |

# 1 Welcome

Welcome to FSK116. Please download this study guide and keep it handy for reference. Undoubtedly, you will be quite familiar with many of the topics covered in the course. This is likely from your everyday life experiences with the concepts or from what you have learnt in school or elsewhere. This is so because Physics is at the heart of everything in nature!

Mathematics is a big part of Physics – it is the language of science. Because you will be practising Physics at an advanced level, the language that will be employed is stricter: the mathematics is generalised and the definitions required are precise. You will thus be treating familiar concepts in general using more powerful formulations and techniques... **Thinking logically is a critical requirement here. You will encounter situations where the final answer to a problem (you are working on) is correct, but because the intervening steps are not logical, full marks are not awarded! How you got to the final answer is much more important than simply getting the correct final answer!**

Lastly, the prescribed textbook is a marvel! It is beautifully written and quite enjoyable to read. So, make it your companion; and don't sell it after passing your first year.

**Studying in the times of Covid-19 is challenging! The teaching and Learning is not traditional! The University of Pretoria is moving towards an “inverted classroom” approach where in the student takes a much more active role in his/her learning. You are required to engage with the work before “coming to class” and to engage with your tutors, peers, and with the lecturer between classes... Most of this will take place virtually, without face-to-face contact. The regulations that govern our response to Covid-19 may change and it may be possible to have face-to-face contact with you... If this happens, we may modify or change how we do things. These changes will be communicated to you primarily through ClickUP, when they happen!**

**I include some FAQ and other pertinent information at the end of this study guide.**

So, buckle up and enjoy the ride!

MJ Legodi

February 2021

## 2 Administrative information

### 2.1 Contact details of the team

**Mr MJ Legodi (Lecturer and course co-ordinator)**

Office 5-5, Natural Sciences 1 Building (NS1)

Email: [matshisa.legodi@up.ac.za](mailto:matshisa.legodi@up.ac.za) (include FSK116 in subject line in all correspondence)

Tel: 012 420 4413 office / 082 458 6918 only for emergencies.

Consulting Hours: Check ClickUP or email for an appointment.

**Mrs L Cilliers (Course Administration)**

Office 4-16, NS1 Building

Tel: 012 420 4268 office

Email: [fsk116.admin@up.ac.za](mailto:fsk116.admin@up.ac.za) or [Laetitia.cilliers@up.ac.za](mailto:Laetitia.cilliers@up.ac.za) (incl. FSK116 in subject field)

**Mrs D Bonner (Course Admin.)**

Office 4-11, NS1 Building

Tel: 012 420 3821 office

Email: [debbie.bonner@up.ac.za](mailto:debbie.bonner@up.ac.za) (incl. FSK116 in subject field)

**Ms Dudu Chinyamakobvu (Practicals co-ordinator)**

First year Labs, 3<sup>rd</sup> Floor, NS1 Building

Email: [dudukelz@gmail.com](mailto:dudukelz@gmail.com) (incl. FSK116 in subject field)

**Dr Pannan Kyesmen (Supervising Tutor)**

Office 5-6, NS1 Building,

Tel 012 420 2684 or +2760 411 8962

Email: [pannan.kyesmen@up.ac.za](mailto:pannan.kyesmen@up.ac.za) (incl. FSK116 in subject field)

**Mr Andre van den Heever (Laboratory Manager and Practical Exemptions)**

Office 4-14, NS1 Building

Tel: 012 420 2094

Email: [andre.vandenheever@up.ac.za](mailto:andre.vandenheever@up.ac.za) (incl FSK116 in subject field)

**Feeling a bit lost...**



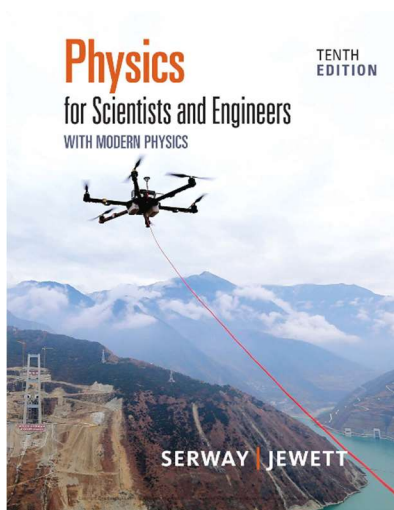
Your Faculty Student Advisor can advise you on goal-setting, adjustment to university life, time management, study methods, stress management and career exploration.

Book an individual consultation or attend a workshop. For other support services see Section 4.

## 2.2 Prescribed Book

It is essential that you have access to a copy of the prescribed textbook as it will be referred to during the lectures and all tutorial problems will be assigned from it.

The prescribed textbook is:



**If you decide to purchase the 9<sup>th</sup> edition, you are responsible for ensuring that the question numbers and content are correlated with those in the 10<sup>th</sup> edition.**

Physics for Scientists and Engineers with Modern Physics,  
10th Edition  
Raymond A. Serway; John W. Jewett

ISBN-10: 1-337-55329-8

ISBN-13: 978-1-337-55329-2

The hardcopy retails for around R 1000 and the eBook around R 650.

Electronic books are also available through companies like VitalSource's [www.Bookshef.com](http://www.Bookshef.com), [www.wizebooks.co.za](http://www.wizebooks.co.za), also at 012 362 5885 and [www.takealot.co.za](http://www.takealot.co.za), etc. Be on the lookout for various marketplaces in the university's environment where you can purchase second-hand books. We do not accept any liability arising from our provision of this information!

The textbook contains a number of *worked out Examples*, some of which will be discussed in the lectures. Carefully work through the *Examples* not covered during the lectures but are in the scope of work described in section 2.10 below.

## 2.3 Mode of Presentation and Online sessions times

**Because of Covid-19, our contact will primarily be online and not face-to-face. However, we endeavour to create contact sessions whenever it is possible, within the constraints of the governments' and university's pandemic response rules.**

**Videos/PDFs of the week's work will be posted on ClickUP. Your duty is to watch these at your own time, but, before the scheduled lectures and to use your textbook and other resources to understand the work. During the scheduled lecture times, the lecturer will go through the work with you, clarify issues and look at some exercises and activities to cement the work.**

**FSK116 comprises of 3 parts: the Theory where you learn Physics principles and concepts, the Practicals where you deepen your understanding of the theory by apply techniques methods to practical problems to demonstrate physics principles and to learn more about them, and, the Tutorials, where you learn how to apply the theory to solve problems.**

**Theory:** There are 4 scheduled lectures per week. The EBIT faculty will allocate you to either of two groups, G01 and G02. Please attend your allocated group lectures through Blackboard Collaborate (BB Collab or BBC). You will find an Online BB Collab link on the FSK116 ClickUP page. You can refer

to the University of Pretoria 2021 Timetable for times.

Attendance and participation during lectures forms an integral part of the course and important information is often disseminated during these times through announcements.

**and Tutorials:** One 3 hour composite session is scheduled for each week. This session is split between a practical session ( first 60 minutes) and a tutorial session (next 120 minutes). See the University of Pretoria 2021 Timetable for times allocated to your specific group. The same BB Collab groups will be used for the practicals and tutorials, so, you do not need to log out of BB Collab. Information on these groups will be posted on ClickUP. Practical and tutorials are compulsory.

You will get a set of Tutorial Problems to prepare for the tutorial session some days before the next tutorial session. Keep your workings / solutions / attempts at these problems in a notebook. No tutor is obligated to help you if you do not present your attempts to them!

These sessions will be recorded and made available to you a few days after the sessions.

## 2.4 ClickUP and Cengage's WebAssign

ClickUP is our primary method of communicating with you. Please make sure that you login to ClickUP several times a week - if not every day. All-important information, such as occasional extra class notes, tutorial and homework problems, marks, practical schedules, practical group allocations, test and examination dates (and venues if we are to write invigilated exams) etc. will be posted on ClickUP. Ensure that you link your preferred email address with ClickUP but also check your allocated UP email address regularly or better still, link this email to your regular email address.

On ClickUP, on the left-hand side menu, there is a link to WebAssign. You will receive most of your tutorial tests, parts of the semester tests and parts of the exams from this site. Access the site only through ClickUP and not directly through a URL on a browser. You will also find a Help Page for WebAssign and importantly, a link on how to enter scientific notation in WebAssign. **WebAssign is strict on how you input numbers and units; spaces and capitals are very important! Your answers will not be interpreted correctly if they are in the wrong syntax. In fact, your answers will not be saved in the system and you will lose marks!**

When you have finished writing an online test, before the final submission, print your answers out as a PDF and save it for your records.

### Final Marks

**Final marks will be published by the EBIT faculty administration** and will be available on your UP Portal page. Do not use "Print/View Academic record" to obtain your results.

The Physics Department may not release any final marks to students. Below are some codes you may encounter when obtaining your final mark.

984 - Did not comply with minimum requirements

987 - Absent from exam

988 - Exam refusal (You are not allowed to sit for the June Exam! This is the end of the end-of-the-road for you!)

989 - Exam result not available

992 - Subminimum (eg 40%) not obtained in exam

999 - Admitted to supplementary exam

## 2.5 Important dates

| Event                                                                  | Date                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lectures Start                                                         | 15 March.                                                                                                                                                                                                                                                                                             |
| Self-Study<br>(High school Revision, Differentiation and Integration). | Monday 15 <sup>th</sup> – Monday 22 <sup>nd</sup> March.                                                                                                                                                                                                                                              |
| Tutorials Start                                                        | 23 <sup>rd</sup> March for G02 / 24 <sup>th</sup> March for G01.<br><br>Tutorial test released on Wednesday 24 <sup>th</sup> March and is due before 18:00 on Thursday 25 <sup>th</sup> March.<br>Subsequent tutorial tests follow in the same vein.                                                  |
| Practicals Start                                                       | 23 <sup>rd</sup> March for G02 / 24 <sup>th</sup> March for G01.<br><br>The first practical will be released by Monday 22 March and is due before 18:00 on Friday 26 March. Subsequent practicals submissions follow in the same vein.<br><br>A Practical Test is written at the end of the semester. |
| Online Tests                                                           | To be announced on ClickUP                                                                                                                                                                                                                                                                            |
| EBIT Test Weeks                                                        | 10, 17 and 19 – 23 April: Semester Test 1<br>05, 07 – 12 June: Semester Test 2                                                                                                                                                                                                                        |
| Practical Test                                                         | 28 June (provisionally)                                                                                                                                                                                                                                                                               |
| Aegrotat (Sick) Test                                                   | 25 June (provisionally)                                                                                                                                                                                                                                                                               |
| Lectures End                                                           | 05 July                                                                                                                                                                                                                                                                                               |
| June Examination                                                       | 12 July (provisionally)                                                                                                                                                                                                                                                                               |
| Weekly Work Schedule                                                   | We will do the <a href="#">work according to this schedule</a>                                                                                                                                                                                                                                        |

## 2.6 Lecture Timetable

| Contact session                          | Day   | Periods | Time          | Venue / Channel |
|------------------------------------------|-------|---------|---------------|-----------------|
| <b>G01 = Study Programme M1R1</b>        |       |         |               |                 |
| Lecture 1                                | Mon   | 5       | 11:30 - 12:20 | BB Collaborate  |
| Lecture 2                                | Tues  | 8       | 14:30 - 15:20 |                 |
| Lecture 3                                | Thurs | 3       | 09:30 - 10:20 |                 |
| Lecture 4                                | Fri   | 6       | 12:30 - 13:20 |                 |
| Tutorial/Practical                       | Wed   | 7-9     | 14:30 - 17:20 | BB Collaborate  |
| <b>G02 = Study Programme b1m1r1 B1C1</b> |       |         |               |                 |
| Lecture 1                                | Mon   | 3       | 09:30 - 10:20 | BB Collaborate  |
| Lecture 2                                | Tues  | 2       | 08:30 - 09:20 |                 |
| Lecture 3                                | Thurs | 1       | 07:30 - 08:20 |                 |
| Lecture 4                                | Fri   | 4       | 10:30 - 11:20 |                 |
| Tutorial/Practical                       | Tues  | 4-6     | 10:30 - 13:20 | BB Collaborate  |

## 2.7 Code of conduct

We are not only facilitating learning in a module, we are also preparing you for the world of work. We expect you to adhere to the code of conduct as spelled out in the Escalation policy of the University of Pretoria.

### 2.7.1 *Communication via email*

When you send an email to your lecturer or any member of the FSK116 team, you have to use a respectful tone and include all of the following aspects:

- A clear, explanatory subject line (e.g. "Submission of FSK116 sick note – P Mdluli, s2145358");
- Your full name and surname at the end of the mail;
- Your student number;
- The module involved; and
- Short and clear message.

### 2.7.2 *Plagiarism*

Plagiarism is a serious form of academic misconduct. It involves both appropriating someone else's work and passing it off as one's own work afterwards. Thus, you commit plagiarism when you present someone else's written or creative work (words, images, ideas, opinions, discoveries, artwork, music, recordings, computer-generated work, etc.) as your own. Only hand in your own original work. Indicate precisely and accurately when you have used information provided by someone else. Referencing must be done in accordance with a recognised system. Indicate whether you have downloaded information from the Internet. For more details, visit the library's website:

<http://www.library.up.ac.za/plagiarism/index.htm>.

**UP's Plagiarism policy strictly forbids assistance of students by anyone/any means during the conduct of tests or exams. We routinely check sites like Chegg for copyrighted UP content and take action against the students who are found to be cheating and dishonest.**

### 2.7.3 *Compliments and complaints*

You are more than welcome to express your appreciation to your lecturer or tutor/demmie and supply feedback about aspects of the course that you enjoy and find valuable.

If you have a query or complaint, you have to submit it in writing with specifics of the issue or the nature of the complaint. It is imperative that you follow the procedure outlined below in order to resolve your issues:

**1. Consult the lecturer concerned about your complaint/concerns.**

If the matter has not been resolved,

**2. Consult the class representative** (The primary function of the Class Representative is to serve as a two-way communication channel between the class and the lecturer).

If the matter has not been resolved,

**3. Consult the module co-ordinator (large modules with multiple lecturers)**

If the matter has not been resolved,

**4. Consult the Head of Department**

If the matter has still not been resolved,

**5. Consult with the Dean of the Faculty**



## 2.8 Main Outcomes

At the end of this module you should be able to apply the principles learnt in the course to analyse and solve relevant problems in: Kinematics, Mechanics, Oscillations, Waves and Thermodynamics. Details for each section can be found in Topics Covered in section 2.10 or on ClickUP.

## 2.9 Learning assumed to be in place

A good grounding in Grade 12 Mathematics is assumed to be in place. To brush up, students are advised to consult the Appendices A, B and D in the prescribed textbook by Serway and Jewett. Another useful source is Lötzt Strauss' Mathematics for Physical Science. Here, the first four chapters are of interest. You do not need to memorise all the formulae or derivations in these sources, but, know and remember the basic ones.

## 2.10 Summary of topics and chapters to be covered

The following summary is based on the prescribed textbook by Serway and Jewett. Please see ClickUP for more details on each topic covered.

| Topic                                        | Chapter           | Main Concepts                                                                                                                           | Study material                                                                                                                |
|----------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Physics and Measurement                      | 1                 | Physical quantities, Models, Dimensional Analysis, Units, Order of magnitude, Significant figures                                       | Ch1 and Appendix A, B, D (Serway and Jewett, 10th edition)<br><br>Ch1 & Ch2 (Lötzt Strauss' Mathematics for Physical Science) |
| Motion in 1 dimension                        | 2                 | Position, Velocity, Acceleration, Kinematic equations under constant acceleration, Motion diagrams                                      | Ch2 (Serway & Jewett)                                                                                                         |
| Vectors                                      | 3                 | Coordinate systems, Vector and scalar quantities, Components of a vector, Unit vectors                                                  | Ch3 (Serway & Jewett)<br>Ch4 (Lötzt Strauss' Mathematics for Physical Science)                                                |
| Motion in 2 dimensions                       | 4                 | Two-dimensional motion with constant acceleration, Projectile motion, Uniform circular motion, Relative velocity and acceleration       | Ch4 (Serway & Jewett)                                                                                                         |
| The laws of Motion                           | 5                 | Force, Newton's laws, Inertia, Mass, Gravitational force and weight, Frictional forces                                                  | Ch5 (Serway & Jewett)                                                                                                         |
| Circular Motion and Resistive Forces         | 6                 | Uniform circular motion (extended), Motion with resistive forces                                                                        | Ch6 (Serway & Jewett)                                                                                                         |
| Energy of a System and Universal gravitation | 7 and 13.1 – 13.2 | Systems and environments, Work done by constant and variable forces, Kinetic energy-work theorem, Potential Energy, Conservative forces | Ch7 & Ch13 (Serway & Jewett)                                                                                                  |
| Conservation of Energy                       | 8                 | Isolated and non-isolated systems, Power                                                                                                | Ch7 & Ch8 (Serway & Jewett)                                                                                                   |
| Linear Motion and Collisions                 | 9                 | Linear momentum for isolated and non-isolated systems, Collisions in 1- and 2-dimensions, Center of mass, Systems of particles          | Ch9 (Serway & Jewett)                                                                                                         |

|                                                                     |             |                                                                                                                                             |                                                                                              |
|---------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Rotation of a rigid object around a fixed axis and Angular Momentum | 10 and 11.1 | Angular position, velocity, acceleration and momentum, Constant angular acceleration, Torque, Moments of inertia, Rotational kinetic energy | Ch10 (Serway & Jewett)<br>Ch2 (Serway & Jewett)<br>Ch11 up to section 11.3 (Serway & Jewett) |
| Oscillatory Motion                                                  | 15          | Motion of objects connected to springs, Simple harmonic motion, Energy of SHM, Pendulum                                                     | Ch15 (Serway & Jewett)                                                                       |
| Wave Motion                                                         | 16          | Propagation of a wave, Traveling waves, Speed of waves on strings, Reflection and transmission                                              | Ch16 (Serway & Jewett)                                                                       |
| Superposition and Standing Waves                                    | 17          | Interference, Standing waves, Boundary conditions, Air columns, Beats                                                                       | Ch17 (Serway & Jewett)                                                                       |
| Temperature                                                         | 18          | Zeroth law, Temperature scales, Thermometers, Thermal expansion of solids and liquids                                                       | Ch18 (Serway & Jewett)                                                                       |
| First Law of Thermodynamics                                         | 19          | Heat, Internal energy, Specific heat, Calorimetry, Latent heat, Work and heat in thermodynamic processes, First law, Applications           | Ch19 (Serway & Jewett)                                                                       |

## 3 Assessment

### 3.1 Assessment Policy

#### 3.1.1 General

Physics cannot be mastered in a few days of “cramming”. You need to practice Physics in order to improve your problem solving skills. The course carries 16 credits, and therefore on average you should spend about 160 hours on this course during the semester. This equates to spending about 9 hours per week on Physics besides the actual lectures - this will include consolidating the theory, preparation for the tutorials (homework problems), and preparation for the practicals.

In order to encourage students to study throughout the semester, a strong emphasis is placed on continuous assessment. For this reason the Semester mark (made up of online tests, tutorials tests, semester tests and practicals conducted during the semester) makes up half of the final mark. Section 3.1.3 below is a summary of the various assessment opportunities and the table shows the weights associated with each, to make up your semester mark.

| Assessment           | Weight      |
|----------------------|-------------|
| Semester Test 1      | 25%         |
| Semester Test 2      | 30%         |
| Online Test Average  | 10%         |
| Weekly Tutorials     | 15%         |
| Weekly Practical     | 12%         |
| Practical Test       | 8%          |
| <b>Semester Mark</b> | <b>100%</b> |

A subminimum of 40% in the Practical Test, 90% attendance of Weekly Practical and a weighted average of 50% in the Practical Mark.

### 3.1.2 Assessment Approach

Assessment will focus mainly on the solving of problems, on a similar level to those given as homework exercises. Marks are allocated for clear, logical reasoning and for appropriate and fully labelled diagrams or sketches.

You need to demonstrate clearly that you understand the physics that you use to answer the question. **This means that a numerically correct answer will not necessarily be awarded full marks.** Knowledge of the theoretical aspects of the course (such as derivations from first principles) will also be assessed. Here, strong emphasis will be placed on clear, logical reasoning.

It is essential that correct units and number of significant digits are given in all answers. Marks are deducted in cases where these are not supplied.

### 3.1.3 Assessment Opportunities

**NO PROGRAMMABLE CALCULATORS MAY BE USED DURING TESTS OR EXAMS.**

- **Online Tests:** The online tests will either be a ClickUP test or WebAssign test. The ClickUP announcement will make it clear which type of test it is. Online tests will take the place of class tests, and will be made available on ClickUP throughout the semester. An announcement will be placed informing you of the period during which the test is available. These tests contribute a significant amount to your Semester Mark.

These tests will be available after a section of work has been completed. The online tests will be available for a couple of days and have a definite, strict opening date and time and closing date and time. These tests can be completed anywhere, so you do not have to do them in the IT labs if you are on campus. However, make sure that you have good internet access.

- **Tutorial Tests:** Tutorial tests will be written at the END of each tutorial session. These tests will be used in the calculation of your Semester Mark.

- **Class Tests:** There will not be class tests, as these have been replaced by the Online Tests.

- **Semester Tests:** These tests will cover a larger amount of work than the Online Tests and will be held during the allocated Test Weeks which are scheduled in the timetable. Times (and venues if they are invigilated face-to-face) will be announced on ClickUP.

- **Practicals:** Each week there is a practical session and each week you will complete a practical report and submit it for marking by the demmie. You may be required to fill in a pre-practical form and present it to the demmie before each practical or as directed.

- **Practical Test:** A practical test will be written at the end of the semester and will cover the work done during the practical sessions. You will be given some experimental data, and will then have to draw graph(s) from which you will have to calculate certain values, quantities and draw some conclusions. You will not have to do the actual measurements. Times (and venues, if these are invigilated) will be announced on ClickUP.

### 3.1.4 Absence from a scheduled Assessment

- **Aegrotat (Sick) Tests:** Should you have been ill for any of the tests listed above, you will need to fill in the form "Absent from Test" on ClickUP under "Forms" (for semester tests) or fill in the "Absent from Tut Test" on your first day back after being booked off, along with a medical certificate. If you do not have a valid medical certificate you will receive zero (0) for that assessment.

**Only medical certificates that comply with the university regulations will be accepted. (Doctors need to have been consulted on the day of the test or a day or two before – NOT THE DAY AFTER THE TEST!!).**

**The following guidelines will be followed:**

**Tutorial Tests:** The average of the remaining tutorial tests will be taken.

**Weekly Practicals:** If you missed a practical, you **MUST** register for and attend one of the catch-up labs (See “Absent from Practical” under “Forms” on ClickUP). **You have to register for these sessions.** Failure to register and attend these sessions will result in you receiving 0 (Zero) for the specific practical, and can jeopardize your exam entrance.

**Online Tests:** There is no sick test for the online tests. A valid doctor’s certificate will ensure that the specific test will not be taken into account for the semester mark. You will still need to fill in the form “Absent from Online Test” (on ClickUP under “Forms”) and send it in on your first day back. During the semester you will need to complete at least one Online Test, or else receive zero for this component of your semester mark.

**Semester Tests:** You will need to fill in the form “Absent from Semester Test” on ClickUP under “Forms” and send it in on your first day back to register for the sick test.

One aegrotat test will be scheduled during the last week of the semester and this will cover ALL the work covered during the semester. This will be the test for all students who have missed a semester test during the semester. Please register your absence from the semester test by sending in the form mentioned above. The date and venue will be confirmed in class and on ClickUP.

**Examination:** Should you be ill for the examination, you have to apply for an aegrotat examination via the **faculty administration not the Physics Department.**

### 3.1.5 Examinations

- **Admission to Examination:**

NB: In order to be admitted to write the examination, the following criteria need to be met:



1. Semester Mark minimum of 40%
2. Final Practical Mark minimum of 50% (Practical reports & Test)
3. Attendance of a minimum of 90% of the practicals and tutorials

Failure to meet **ANY** of these criteria will result in refusal to write the examination. Please see the regulation Eng. 3 (b) in the Yearbook of the Engineering Faculty.

- **Aegrotat (Sick) and Supplementary Examinations:**

**Aegrotat (Sick) Examination:** Should you have been ill for the examination, you have to apply for an aegrotat (sick) examination via the faculty administration.

**Supplementary examinations** are granted automatically when:



1. a Final Mark of between 45% and 49% is achieved
2. a Final Mark  $\geq 50\%$  achieved BUT the required subminimum of 40% was not achieved in the examination
3. a Final Mark of between 40% and 44% was achieved and where the candidate ALSO achieved either a semester mark or an examination mark of 50% or higher

The supplementary exam will be written on the date published by the University in the supplementary examinations timetable.

**NB:** *The maximum final mark for the course which can be awarded after supplementary examination is 50%.*

Please see the regulation Eng. 3 (e) in the Yearbook of the Engineering Faculty.

### 3.1.6 *Cheat Sheets\**

During semester tests and exams, NO FORMULA sheet will be handed out with the question paper. It will be your responsibility to draw up your own formula sheet — your Cheat Sheet.

### 3.1.7 *Rules for Cheat Sheets*



- Read the instructions at the top of the Cheat Sheet.
- It must be on the official Cheat Sheet for the specific test/exam printed on A4.
- It must be hand written.
- It must be written in colour – any colour but NOT BLACK.
- It may NOT have derivations or worked out solutions to specific problems.
- It must be handed in with your test. Should you want it back afterwards, please staple it to your paper (with your own staple and stapler).
- The page may not be folded.
- You will receive an additional 2% for handing in your cheat sheet **and** having your barcode sticker - see the item “Barcodes” below. If you do not, however, hand in your cheat sheet, you will not receive the additional marks.
- You may only have ONE cheat sheet in the exam/test venue.
- You may only write on one side of the page for a semester test, but for the sick test or the examination, you may use both sides of the page.

\*Students are allowed to bring and use Cheat Sheets during invigilated tests or exams. Students can compile these and use them during online tests or exams. They are a handy reference!

### 3.1.8 *Pass Mark*

**NB:** Even if a student obtains 50% or more for the Final Mark, BUT obtains less than 40% for the examination, that student will not pass the module, but will be granted a supplementary examination.

**NB:** A student will pass the module with distinction if the final mark is  $\geq 75\%$ . Please see the regulation Eng. 3 (c) in the Yearbook of the Engineering Faculty.

### 3.1.9 *Grading and administrative errors*

When your tests are returned to you, please ensure that all the questions have been marked, that the marks allocated on your test are correct and that the **correct marks** appear on ClickUP. Should there be any unmarked questions, these must be brought to the attention of Mr Legodi **immediately**. Also, should there be any other discrepancies, please complete the form “Semester Test re-mark” (on ClickUP under “Forms”) within 7 days of the graded papers being released. **NO CHANGES WILL BE MADE TO PAPERS RECEIVED LATER THAN ONE WEEK AFTER THE GRADED PAPERS HAVE BEEN HANDED OUT.**

### 3.1.10 Practical component of the course

#### ATTENDANCE OF PRACTICAL SESSIONS IS COMPULSORY

The allocated groups will be announced on ClickUP. Please consult Ms Dudu Chinyamakobvu or Mrs Cilliers with regards to problems in practical time allocations.

**NB:** Inform Ms Dudu Chinyamakobvu or Mrs Cilliers immediately if you are not on the practical list.

#### EVERYONE has to attend the practical sessions.

You have to attend all practical sessions. If you have missed a session with a valid excuse (there are not many of these!), you need to register for a catch-up lab with the practical co-ordinators, Ms Dudu Chinyamakobvu. This applies to late registrations also!

Students repeating the FSK176 / FSK116 course, **may apply for exemption** from the practical course by filling in the “Exemption from Practicals” form (on ClickUP under “Forms”). Successful applicants will be notified, but until such notification has been received, you need to attend the practicals.

**Please note that successful applicants still have to attend the compulsory tutorial sessions.**

**Please note: A practical mark of at least 50% and attendance of at least 90% of practical sessions are required for examination entrance.**

- **Practical Marks:** After completion of the experiment, the assistant will mark the work you have submitted in your practical report.
- **Practical Test:** Towards the end of the semester – usually in the last week or so – you will be given a Practical Test on the practicals done during the semester.

### 3.1.11 Homework and Tutorial Sessions

Tutorial sessions are compulsory for **ALL** students. **No exemption is given for the tutorials.**

The marks received for the tutorial tests all contribute to your Semester Mark and Final Mark.

During the week, you will receive a Tutorial Problem set as homework, which will be similar to the problems which must be solved during the weekly tutorial test. Tutorial tests will be written by all after the Wednesday tutorial session. Details about the Tutorial Tests will be announced on ClickUP.

During the tutorial session, the tutor will be available to guide and assist students with the tutorial problems. This does not mean that the tutor will be providing all the solutions to the problems; the problems are there for students to solve, and the tutor is merely a guide and to hone your problem solving techniques. **DO NOT CONSULT SOLUTIONS BEFORE YOU HAVE ATTEMPTED THE PROBLEM. THIS WILL NOT IMPROVE YOUR PROBLEM SOLVING SKILLS.**

## 3.2 Calculation of the Final Mark (Pass Mark)

| Assessment                                               | Weighting   |
|----------------------------------------------------------|-------------|
| Semester Test 1                                          | 25%         |
| Semester Test 2                                          | 30%         |
| Online Test Average                                      | 10%         |
| Weekly Tutorials                                         | 15%         |
| Weekly Practicals                                        | 12%         |
| Practical Test                                           | 8%          |
| <b>Semester Mark</b>                                     | <b>100%</b> |
| <b>Final Mark</b> $\equiv$ (SEMESTER MARK + EXAM MARK)/2 |             |

**Exemption from practicals:** If you have practical exemption, the calculation of your semester mark is the same as above, but the total mark (now out of 80!) will be converted to a percentage.

## 4 Support services

Please download a QR code reader on your smartphone. To download a QR code reader open your mobile app store (App Store, Google Play or Windows Marketplace) and search for QR code readers.

### 4.1 EBIT FAQ New Students

Link to [FAQ for new students](#).

### 4.2 EBIT FAQ Existing Students

Link to [FAQ for existing students](#).

### 4.3 Safety in the evening and emergencies

- This is applicable when students are on campus.
- For any safety or emergency related matters, eg if you need a security officer to accompany you from your residence to campus, phone the Operational Management Centre (details at the back of your student card).
- The 24-hour, multi-disciplinary UP Crisis Line offers professional and confidential support to victims of crime in times of trauma. For assistance and immediate action, phone the UP Crisis Line on: 0800 00 64 28.
- Hatfield residence students: From 18:00 till 06:00 security officers are available to escort you (on foot) to and from your residence or campus anywhere east of the Hatfield Campus through to the Hillcrest Campus.

### 4.4 E-learning support

- Report a problem you experience to the Student Help Desk on your campus.
- Visit the open labs in the Informatorium Building or IT labs on your campus to report problems at the offices of the Student Help Desk.
- Approach the assistants at the help desks—campus specific (for example: adjacent to the Student Computer Laboratories in IT Building, NW2, CBT or Aldoel Building IT labs, etc).
- For Student Help Desk, Call 012 420 3837.
- Email [studenthelp@up.ac.za](mailto:studenthelp@up.ac.za)

### 4.5 Other support services:

|                                        |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                |                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| FLY@UP:<br>The Finish<br>Line is Yours | <ul style="list-style-type: none"><li>• Think carefully before dropping modules (after the closing date for amendments or cancellation of modules).</li><li>• Make responsible choices with your time and work consistently.</li><li>• Aim for a good semester mark. Don't rely on the examination to pass.</li></ul> | <a href="http://www.up.ac.za/fly@up">www.up.ac.za/fly@up</a><br>email: <a href="mailto:fly@up.ac.za">fly@up.ac.za</a>                                          |  |
| Disability<br>Unit                     | Academic support for students with learning disabilities: <ul style="list-style-type: none"><li>• Assistive technological services</li></ul>                                                                                                                                                                          | <a href="https://www.up.ac.za/disability-unit">https://www.up.ac.za/disability-unit</a><br>012 420 2064<br>email: <a href="mailto:du@up.ac.za">du@up.ac.za</a> |  |

|                                         |                                                                                                                                                                                                                                                                                    |                                                                                      |                                                                                       |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                         | <ul style="list-style-type: none"> <li>• Facilitation of test and examination accommodations</li> <li>• Test and exam concession applications</li> <li>• Accessible study venues and a computer lab</li> <li>• Referrals for recommended textbooks in electronic format</li> </ul> |                                                                                      |                                                                                       |
| Student Counselling Unit                | Provides counselling and therapeutic support to students                                                                                                                                                                                                                           | 012 420 2333                                                                         |    |
| Student Health Services                 | Promotes and assists students with health and wellness                                                                                                                                                                                                                             | 012 420 5233<br>012 420 3423                                                         |    |
| The Careers Office                      | Provides support for UP students and graduates as they prepare for their careers                                                                                                                                                                                                   | <a href="mailto:careerservices@up.ac.za">careerservices@up.ac.za</a><br>012 420 2315 |   |
| Department of Security Services         | 24-hour Operational Management Centre<br>24-hour Operational Manager Crisis Line                                                                                                                                                                                                   | 012 420-2310<br>012 420-2760<br>083 654 0476<br>0800 006 428                         |  |
| Department of Student Affairs           | Enquiries concerning studies, accommodation, food, funds, social activities and personal problems                                                                                                                                                                                  | 012 420 2371/4001<br>Roosmaryn Building, Hatfield campus                             |  |
| Centre for Sexualities, AIDS and Gender | Identifies and provides training of student peer counsellors                                                                                                                                                                                                                       | 012 420 4391                                                                         |  |
| Fees and funding                        | <a href="http://www.up.ac.za/enquiry">http://www.up.ac.za/enquiry</a><br><a href="http://www.up.ac.za/fees-and-funding">www.up.ac.za/fees-and-funding</a>                                                                                                                          | 012 420 3111                                                                         |  |
| IT Helpdesk                             | For student IT related queries                                                                                                                                                                                                                                                     | 012 420 3051<br><a href="mailto:studenthelp@up.ac.za">studenthelp@up.ac.za</a>       |  |



